Childcare Costs & Female Labor Force Participation

DSAN 5650: Causal Inference for Computational Social Science

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A substantial body of research has found that reductions in childcare costs are positively associated with increased maternal labor-force participation. However, when analyzing the elasticity of employment to childcare prices, researchers find heterogeneity across various geographies and time-periods. This may be attributable to differences in educational attainment, labor force attachment, and other factors (Morrissey 2016). This research paper uses U.S. county-level data to examine whether the association between childcare affordability and maternal labor-force participation among women with children under age six varies across local occupational contexts. I present a Bayesian logit-normal regression model that estimates maternal labor force participation as a function of childcare affordability, the county share of workers in "greedy" occupations, educational attainment, geographic rootedness, and an interaction between childcare burden and occupational structure. Aligning with existing research, I find that higher childcare burden is associated with lower maternal labor force participation. However, I also find that this negative association becomes weaker as a county's concentration of women in greedy occupations increases. Educational attainment and geographic rootedness are also moderately associated with maternal labor force participation, while greedy-occupation-share is weakly associated.

Introduction

Earlier this year, I examined the major factors that help explain U.S. maternal labor force participation trends and persistent wage differentials between men and women. Existing research consistently identifies motherhood as an important contributor to both reduced labor-force participation and long-term earnings penalties.

Childcare costs, which have significantly outpaced overall inflation, play a significant role in women's ability or decision to remain in the workforce. This trend has driven an overall focus within the research community on the relationship between childcare affordability and maternal labor force participation. Research has appropriately paid particular attention to low-income women, for whom childcare expenses represent a large share of potential earnings, and whose labor market attachment is the most impacted by childcare costs (https://www.jstor.org/stable/41238095).

Several studies have concluded that reducing childcare costs would increase maternal employment, especially among low-income women. The majority of U.S. childcare assistance programs are also primarily available for lower-income households and reach only a minority of eligible families. Less is known about this association amongst women in moderate or high income brackets, who remain burdened by high childcare costs. For these women, childcare constraints may impact labor force participation differently. Increased earning potential and financial resources may make them more likely to remain employed despite high childcare costs, while reducing hours or seeking work flexibility to meet childcare demands. This behavior may ultimately result in reduced career advancement and lower wages.

This paper tests this theory, using county-level data to analyze how women in different occupational contexts may respond differently to childcare costs. I use the concentration of women working in "greedy" occupations, or high-paying occupations that disproportionately reward long hours and constant availability, to measure local occupational context. This is used to test whether the known association between childcare burden and maternal labor-force participation is moderated by occupation "greediness".

County-level female labor force participation only determines whether mothers participate in the labor force, rather than how many hours they work. Therefore, this analysis does not directly model the wage gap. However, beginning to understand potential effect heterogeneity in this relationship may help explain persistent gender gaps in earnings despite comparable levels of overall labor-force participation.

Background and Conceptual Framework

Existing Childcare Policy and Maternal Employment

Researchers generally agree that increased investment in childcare affordability interventions results in higher maternal labor force participation, particularly among low-income women in the first few years after birth. For instance, Burgess, Chien and Enchautegui find that increasing Child Care and Development Block Grant (CCDBG) expenditures by 10-percent per child would increase employment levels among low-income women with young children by between half to two-thirds of a percent (https://aspe.hhs.gov/sites/default/files/migrated_legacy_files//171051/EffectsCCSubsidiesMaternalLFPBrief

Still, the majority of U.S. childcare assistance programs reach only a subset of families and are concentrated among lower-income households. The Child Care and Development Block Grant (CCDBG) is only available for families who make less than 85% of their state median income.

Even amongst those that meet requirements, federal childcare subsidies only reach approximately 14% of eligible families (https://aspe.hhs.gov/reports/factsheet-estimates-child-care-eligibility-receipt-fiscal-year-2017), and state subsidies reach only approximately 22% of eligible families. Tax incentives, including the Child and Dependent Care Tax Credit, also benefit a minority of families with children due to its non-refundable structure, steep income phase-downs, and restrictions that require two working parents. (https://www2.census.gov/library/working-papers/2025/adrm/ces/CES-WP-25-25.pdf)

While most childcare programs focus on low-income families, evidence from universal early childcare programs suggests that reducing childcare constraints can affect maternal employment across a broader range of women. Two years after implementing a universal pre-kindergarten program in 2008, D.C. observed a 12% increase in the maternal labor force participation rate. A study from the Center for American Progress found that this increase was strongly associated with implementation of universal preschool (https://www.americanprogress.org/article/effects-universal-preschool-washington-d-c/). This example is particularly important given D.C.'s occupational profile, with an unusually large concentration of highly-educated women in high-paying, senior positions. This example motivates the current analysis, which seeks to understand the varying slope of the relationship between childcare affordability and labor force participation among counties with different female occupational profiles.

Occupational "Greediness"

This analysis builds on existing research from economist Claudia Goldin, who has extensively studied "greedy occupations". Goldin finds that greedy occupations lead women to remain employed, even as they have children, but earn less over time, due to childcare responsibilities that prevent them from meeting demanding work requirements, including long hours and travel (Goldin 2014). Women's role as primary care-givers leads them to take career breaks, reduce hours, and trade high-stress roles for flexible roles.

Goldin identifies finance, law, consulting, and corporate management as common greedy occupations, characterized by time inflexibility, low work substitutability, and non-linear pay. These occupations drive gender pay differentials, since women are more likely to experience childcare related work interruptions and less likely to be able to work rigid schedules (Sobeck 2024). Using Goldin's research as a framework, I model county-level occupational greediness by sex using U.S. Census adult employment statistics.

Educational Attainment

Educational attainment is closely related to occupation context and labor force participation. Geographies with a greater share of individuals with advanced degrees are more likely to have higher concentrations of individuals in advanced white-collar occupations. Women in these occupations are more likely to have greater earning power, which also influences their labor force attachment. I model educational attainment as the share of individuals in each county with at least a bachelor's degree. This is used to determine if the effect of greedy occupation structure is partially or completely explained by county educational composition.

Geographic "Rootedness" and Informal Childcare Support

Informal childcare support, through family-members, friends and neighbors, can provide mothers with relief from exorbitant childcare costs, as well as unmatched flexibility. However, the availability of local support varies across households. Highly-educated and high-income individuals are often more likely to move away from their birthplace, where family resides, to regions with greater economic opportunities for higher-paying jobs. While over half of U.S. adults report living within an hour of family, a Pew Research study found that upper-income adults and adults with a postgraduate degree were the most likely to report living near no extended family. https://www.pewresearch.org/short-reads/2022/05/18/more-than-half-of-americans-live-within-an-hour-of-extended-family/

Existing literature suggests that access to informal familial childcare can influence maternal labor force participation. Compton and Pollak find that married women with young children that live within close proximity of mothers or mothers-in-law are predicted to have a 4 to 10 percent higher labor force participation rate when compared to women that live farther from family (https://www.sciencedirect.com/science/article/pii/S0094119013000314). They theorize that family members within close proximity provide "insurance" to working mothers, by being available to provide care in irregular or unanticipated circumstances. Building upon this framework, I use geographic rootedness as a proxy for access to local familial childcare. I model geographic "rootedness" as the share of adults 25 or older in each county who report living in the state they were born in. While this does not directly estimate the proximity of family or availability of informal childcare, it provides a county-level profile of geographic attachment to measure potential differences in access to informal childcare support.

Data and Measures

Data Sources

This analysis uses 2022 data from the Department of Labor's National Database of Childcare Prices (NDCP) to retrieve each county's maternal female labor force participation rate (FLPR) for women with children under 6, the outcome variable, and childcare costs, which are used to determine the childcare cost burden. The American Community Survey (ACS) five-year estimates are used to identify the median family income, share of bachelor's-plus individuals, female greedy occupation share, and geographic rootedness. This county level data is merged using FIPS-codes, resulting in a final sample of 2,653 counties. 491 counties are excluded from the sample as a result of missing any of the modeled variables. Exclusion is primarily a result of missing childcare data, including complete state absences in Pennsylvania, Indiana, Vermont, and New Mexico, due to low response rates or too few childcare providers. This is a known limitation of the NDCP.

Data Transformations

FLPR is first converted from a percentage to a proportion. Exact 0 and 1 FLPR values are then slightly adjusted, since the logit function is undefined at those exact values. Greedy occupation share is represented by the proportion of eligible employed women working in greedy occupations. Bachelor's-plus share is calculated as the proportion of adults age 25 and older with at least a bachelor's degree. Finally, geographic rootedness is calculated as the proportion of adults age 25 and older who were born in their current state of residence. All continuous predictors are standardized to have mean zero and standard deviation prior to modeling.

Data and Measures Summary

The sources and representations of each variable are summarized below:

Variable	Source(s)	Representation
Maternal Female Labor Force Participation Rate (FLPR)	Department of Labor (DOL) county data	Labor-force participation rate among women with children under age six
Childcare cost burden	DOL National Database of Childcare Prices; American Community Survey (ACS) median family income	Annual center-based childcare cost for children ages 0–5 divided by median family income
Greedy occupation share	ACS Occupation by Sex for the Civilian Employed Population 16 Years and Over (Series S240)	Share of employed women in management, business and financial, and legal occupations
Educational attainment	ACS Place of Birth by Educational Attainment in the United States (Series B06009)	Share of adults age 25 or older with a bachelor's, graduate, or professional degree
Geographic rootedness	ACS Place of Birth by Educational Attainment in the United States (Series B06009)	Share of adults age 25 or older who were born in their current state of residence